# Lecture 6 Homework

## Programming

None

## Exercises

1. Show that $x \equiv 2 \bmod 6$ and $x \equiv 3 \bmod 4$ have no simultaneous solutions.
2. Show that $x \equiv 2 \bmod 6$ and $x \equiv 0 \bmod 4$ do have simultaneous solutions.
3. Find an integer $x$ such that $x \equiv a \bmod 12$ and $x \equiv 8 \bmod 35$.
4. Find an integer $x$ such that $3x \equiv 2 \bmod 5$ and $4x \equiv 5 \bmod 7$.
5. Explain why there are $8$ different square roots of $1 \bmod 3 \cdot 5 \cdot 7 = 1 \bmod 105$.
6. Is $2$ a square modulo $101$?
7. Find the principal square root of $2$ modulo $19$.
8. Find the principal square root of $6$ modulo $19$.
9. Find the principal square root of $892$ modulo $1039$.
10. Find a cube root of $2$ modulo $101$. That is, find a cube root of $\overline{2}$ in $Z_{101}^{\times}$.
11. Find a cube root of $46$ modulo $101$. That is, find a cube root of $\overline{46}$ in $Z_{101}^{\times}$.
12. Find a cube root of $61$ modulo $103$. That is, find a cube root of $\overline{61}$ in $Z_{103}^{\times}$.
13. Solve $x^2 \equiv -1 \bmod 13 \cdot 17 \cdot 29$.
14. Find an integer $x$ such that $x \equiv 2 \bmod 3$, $x \equiv 3 \bmod 5$ and $x \equiv 4 \bmod 7$.

## Suggested Reading

- Garrett – Sections 12.6, 13.1-13.3, 13.8
- Smart – Sections 7.3, 11.3, , 17.3.2, 18.2.1
- Forouzan – Sections 1.3.2